7/4/2026

3. Solve $(D^2-6D+9)y = \dfrac{e^{3x}}{x^2}$ by the MVP.

Sol^n: Aux eq^n is:

$$D^2-6D+9=0$$

$\Rightarrow D = 3, 3.$

$\therefore$ C.F is:

$$y_c = (c_1 + c_2 x)e^{3x}$$

$$\Rightarrow y_c = c_1 e^{3x} + c_2 x e^{3x}$$

form: $y_c = c_1 y_1(x) + c_2 y_2(x)$

$\Rightarrow y_1(x) = e^{3x}$; $y_2(x) = x e^{3x}$

Let the P.I

$$y_p = u(x)y_1(x) + v(x)y_2(x)$$

Where

$$u(x) = -\int \frac{y_2(x)\cdot Q(x)}{W}\,dx$$

$$\& \; v(x) = \int \frac{y_1(x)\,Q(x)}{W}\,dx$$

Where $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{3x} & xe^{3x} \\ 3e^{3x} & 3xe^{3x}+e^{3x} \end{vmatrix}$

$$\Rightarrow \boxed{W = 3xe^{6x} + e^{6x} - 3xe^{6x} = e^{6x}}$$

$$\& \; \boxed{Q(x) = \frac{e^{3x}}{x^2}}$$

$$\therefore u(x) = -\int \frac{x\cancel{e^{3x}}\cdot \cancel{e^{3x}}}{x^2\cdot \cancel{e^{6x}}}\,dx$$

$$= -\int \frac{1}{x}\,dx$$

$$\Rightarrow u(x) = -\log x$$

$$v(x) = \int \frac{e^{3x}\cdot e^{3x}}{x^2\cdot e^{6x}}\,dx$$

$$= \int \frac{1}{x^2}\,dx$$

$$= -\frac{1}{x}$$

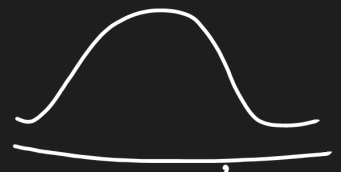
$$\therefore \text{P.I} = y_p = -\log x\, e^{3x} - \frac{1}{x}\, x\cdot e^{3x}$$

Gen. Sol^n is:

$$y = y_c + y_p$$

Module 3: Gamma & Beta fns

Normal distⁿ



$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$\Gamma(k) = \int_0^\infty e^{-x} x^{k-1}\,dx$$

$k > 0$

$$= 2\int_0^\infty e^{-x^2} x^{2k-1}\,dx$$

$\Gamma \to$ Gamma

$\gamma \to$ gamma

$$\beta(p, q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx$$

Γ

<u>Gamma function</u>: For $k>0$,

we define the Gamma function as:

$$\Gamma(k) = \int_0^\infty e^{-x}\, x^{k-1}\, dx.$$

↳ Gamma f^n of the first kind

$$\int_0^\infty e^{-x}\, x^{-\frac{1}{4}}\, dx = \Gamma\left(\tfrac{3}{4}\right)$$

$k-1 = -\frac{1}{4}$

$\Rightarrow k = 3/4$

$$\Gamma\left(\tfrac{1}{2}\right) = \int_0^\infty e^{-x}\, x^{-\frac{1}{2}}\, dx$$

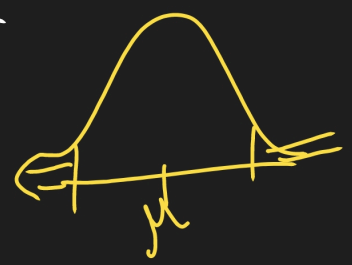
$$\Gamma(5) = \int_0^\infty e^{-x}\, x^{4}\, dx$$

<u>Eg</u>: Write $\Gamma\left(\frac{3}{2}\right)$, $\Gamma\left(\frac{1}{2}\right)$, $\Gamma(5)$

$$\Gamma\left(\tfrac{3}{2}\right) = \int_0^\infty e^{-x}\, x^{\frac{3}{2}-1}\, dx$$

$k = \frac{3}{2}$

$$= \int_0^\infty e^{-x}\, x^{\frac{1}{2}}\, dx$$



<u>Note</u>: we have

$$\Gamma(k) = \int_0^\infty e^{-x} x^{k-1} \, dx$$

In the above integral,

put $x = t^2$

$dx = 2t \, dt$

	x	0	∞
$\sqrt{x}$	$= t$	0	∞

$$\therefore \Gamma(k) = \int_0^\infty e^{-t^2} \cdot (t^2)^{k-1} \cdot 2t \, dt$$

$$= 2 \int_0^\infty e^{-t^2} t^{2k-2+1} \, dt$$

$$\Rightarrow \Gamma(k) = 2 \int_0^\infty e^{-t^2} t^{2k-1} \, dt \quad — (*)$$

$(*)$ is known as gamma f^n of the second kind.

Properties of gamma fn

1. $\Gamma(1) = 1$

Proof: We have

$$\Gamma(k) = \int_0^\infty e^{-x} x^{k-1} dx$$

$$\Rightarrow \Gamma(1) = \int_0^\infty e^{-x} x^{1-1} dx$$

$\boxed{k=1}$

$$\Rightarrow \Gamma(1) = \int_0^\infty e^{-x} x^0 dx$$

$$\Rightarrow \Gamma(1) = \int_0^\infty e^{-x} dx$$

$$= \left.\frac{e^{-x}}{-1}\right]_0^\infty$$

$$= -1[0-1]$$

$$\Rightarrow \Gamma(1) = 1$$

2. $\Gamma(n+1) = n\Gamma(n)$

$(\text{or } \Gamma(n) = (n-1)\Gamma(n-1))$

Proof: $\Gamma(n+1) = \int_0^\infty e^{-x} \underbrace{x^n}_{u} dx$

Integrating by parts

$$\Gamma(n+1) = \int_0^\infty x^n d(-e^{-x})$$

$$= x^n(-e^{-x})\Big]_0^\infty - \int_0^\infty -e^{-x} n x^{n-1} dx$$

$$\left[\int u\, dv = uv - \int v\, du\right]$$

$$\Rightarrow \Gamma(n+1) = 0 + n\int_0^\infty e^{-x} \underbrace{x^{n-1}}_{\Gamma(n)} dx$$

$$\int e^{-x} x^n dx$$

$$= \int x^n \cdot d(-e^{-x})$$

$$d(-e^{-x}) = -d(e^{-x})$$

$$= -[-e^{-x}]dx$$

$$= e^{-x} dx$$

$$\Gamma(n+1) = \int_0^\infty e^{-x} \underset{\substack{\downarrow \\ u}}{x^n} \, dx$$

u dv

$$= x^n(-e^{-x})\Big]_0^\infty - \int_0^\infty (-e^{-x}) \cdot n \cdot x^{n-1} \, dx$$

$$= 0 + n\int_0^\infty e^{-x} x^{n-1} \, dx$$

$$\Rightarrow \Gamma(n+1) = n\Gamma(n)$$